Technical Document: Demystifying important messages found in the message.log file & how to configure Liberty so these messages will be written to the z/OS operator console

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Overview

This document was written by the WSC to help users understand the variety of different message IDs you may find within your message.log files for your z/OS Connect servers. We have included messages that we have repeatedly seen from our own experiences. Each section contains a short description, various message IDs, the corresponding base message, a link to IBM documentation to learn more, and a coding sample that displays the configuration element to enable the specific message IDs to be written to the z/OS operator console.

There are two distinct main sections of this document, z/OS Connect specific messages and Liberty specific messages. For z/OS Connect, we have broken down the sub-sections by message prefix (e.g. BAQR) and then by corresponding z/OS subsystem (e.g. CICS) or feature (e.g. data transformation, Gradle plugin). We have color-coded the message IDs that are specific to z/OS Connect version 2.0 and 3.0:

Messages highlighted with are specific to z/OS Connect 2.0

Messages highlighted with are specific to z/OS Connect 3.0

For Liberty, we have broken down the sub-sections by message prefix (e.g. SRVE). There are many message codes for Liberty that do not pertain to z/OS Connect or z/OS Connect servers, we added the ones we feel are most beneficial to be aware of for users.

The Anatomy of a Message.log Message ID is broken down below:

1. Prefix – Identifies the corresponding product family
2. Sub-component – Identifies the internal module or function
3. Message Number – A unique 4-digit serial number to distinguish the exact error/event
4. Severity code – indicates the priority level of the log entry (I – Information, W – Warning, E – Error, S – Severe)

For example, **BAQR0657E** → BAQ (Prefix), R (Sub-component), 0657 (Message number), E (Severity code).

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Notes & Disclaimers

This technical document is not meant to be a comprehensive list of all possible information, warning, and error messages that you may encounter. There are a vast range of message IDs that are not included in this technical document. As stated in the overview, we have listed IDs that the WSC team has seen through our own experiences and identified as beneficial to share within this document.

This document has a live table of contents, click on the section you would like to browse, and you will be taken straight there. There is a return to Table of Contents link at the top of each page to allow for easier navigation of the document.

If you have any questions or feedback, reach out to the emails listed in the footer.

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z/OS Connect Messages

BAQR Messages

BAQR Prefix Messages can produce a wide array of messages pertaining to core functions of z/OS Connect, CICS, & Db2.

z/OS Connect

Messages produced by the core functions of IBM z/OS Connect

- **BAQR0429W**: API {0} encountered an error while processing a request under URL {1}. (IBM Docs)
- **BAQR7039E**: z/OS Connect EE API archive file location {0} does not exist. (IBM Docs)
- **BAQR7042E**: z/OS Connect EE service archive file location {0} does not exist. (IBM Docs)
- **BAQR1108E**: z/OS Connect API requester archive file location {0} does not exist. (IBM Docs)
- **BAQR0558E**: The remote service invocation failed with response message: {0} and response body: {1}. (IBM Docs)

Coding sample to enable core z/OS Connect function messages to MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="BAQR0429W, BAQR7039E, BAQR7042E, BAQR1108E, BAQR0558E"/>

CICS

Messages codes related to CICS interactions with z/OS Connect

- **BAQR0657E**: Transaction abend {0} occurred in CICS while using CICS connection {1} and service {2}. The supplied code was returned by CICS (IBM Docs)
- **BAQR0658E**: A security error occurred in CICS while using CICS connection {0} and service {1}. (IBM Docs)
- **BAQR0686E**: Program {0} is not available in the CICS region with connection ID {1}; service {2} failed. (IBM Docs)
- **BAQR0687E**: Request failed with a CICS NOTAUTH error while using CICS connection {0} and service {1} (IBM Docs)
- **BAQR0660E**: Failed to establish CICS connection {0} to host {1} on port {2}, reason: {3}. (IBM Docs)

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- **BAQR0661E**: A connectivity failure occurred while using CICS connection {0} and service {1}. (IBM Docs)
- **BAQR0662E**: No free sessions were available on CICS connection {0}. (IBM Docs)
- **BAQR0663E**: CICS connection {0}, failed to establish. (IBM Docs)
- **BAQR0664E**: Communication failure for CICS connection {0} to host {1} on port {2}, reason: {3}. (IBM Docs)

  Coding sample to enable CICS messages to MVS Log:
  <zosLogging enableLogToMVS="true" wtoMessage="BAQR0657E, BAQR0658E, BAQR0686E, BAQR0687E, BAQR0660E"/>

Db2

Messages codes related to Db2 interactions with z/OS Connect

- **BAQR1507E**: Error invoking endpoint: {0} (IBM Docs)
- **BAQR1508E**: Error parsing Db2 response: {0} (IBM Docs)
- **BAQR1509E**: Error calling Db2 REST service with HTTP response code: {0} and response body: {1} (IBM Docs)

  Coding sample to enable Db2 messages to MVS Log:
  <zosLogging enableLogToMVS="true"
  wtoMessage="BAQR1507E, BAQR1508E, BAQR1509E"/>

BAQG Messages

BAQG Prefix Messages produce messages pertaining to the Gradle plugin for z/OS Connect

- **BAQG0003E:** openApiGenerate task failed since inputSpec does not exist (IBM Docs).
  - Most common startup failure. May be caused by incorrect path to build.gradle.
- **BAQG0009E:** Could not parse file "{0}". Exception message: {1} (IBM Docs).
  - An error occurred while processing the file. See the exception message for more information.

  Coding sample to enable Gradle plugin messages to MVS Log:
  <zosLogging enableLogToMVS="true"
  wtoMessage="BAQG0003E, BAQG0009E"/>

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BAQW Messages

BAQG Prefix Messages produce messages pertaining to z/OS Connect Designer

- **BAQW0001E:** Error parsing OpenAPI document: {0} (IBM Docs).
    - Check for extra messages that explain the exact issue.
- **BAQW0012E:** An error occurred while attempting to load JSON schema {0}: {1} (IBM Docs).
    - Use the specified error to determine the cause of the error loading the schema.
- **BAQW0070E:** The API project file build.gradle does not exist in the expected location {0} build.gradle. (IBM Docs)

Coding sample to enable z/OS Connect Designer messages to MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="BAQW0001E, BAQW0012E, BAQW0070E"/>

GMO Messages

IMS

GMOx Prefix Messages are produced by the IMS service provider

- **GMOBA0113E**: The IMS gateway server was unable to process the request because an unexpected error occurred. (IBM Docs)
- **GMOBA0108E**: The IMS gateway server is unable to establish a connection with IMS Connect: Failed to connect to host [host_name], port [port_number]. [java_exception] (IBM Docs)

Coding sample to enable IMS service provider messages to MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="GMOBA0113E, GMOBA0108E "/>

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Data transformation

GMOM Prefix Messages are produced by the data transformation function that converts between byte arrays and JSON

- **GMOMW0005E**: A data type conversion error occurred while the leaf field {field_name} of service interface {service_interface_name} was converted for {service name: [service_name]:[error_details]} or a data type conversion error occurred while the leaf field {field_name} of service interface {service_interface_name} was converted for {[service_name], [api_name], [api_operation]: [error_details]}. (IBM Docs)
- **GMOMW0011E**: The maximum of {max_number_of_entries} entries for the leaf array {field_name} of service interface {service_interface_name} was exceeded. The number of the entries for the leaf array exceeds what was expected. (IBM Docs)

Coding sample to enable data transformation messages to MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="GMOMW0005E, GMOMW0011E"/>

Liberty messages

SRVE Messages

SRVE prefix messages are servlet engine log codes indicating web module lifecycle events, warnings, or runtime errors (IBM Docs)

- **SRVE0011E**: An illegal argument exception occurred during buffer output
- **SRVE0169I**: A web module is loading.
- **SRVE0250I**: Web Module {0} has been bound to {1}.
- **SRVE0232E**: An internal server error occurred while processing a request.

Coding sample to enable servlet engine log messages:
<zosLogging enableLogToMVS="true"
wtoMessage="SRVE0011E, SRVE0169I, SRVE0250I, SRVE0232E"/>

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CWWKx Messages

CWWKx prefix groups cover Liberty core kernel operations, application lifecycle management, security, web container routing or startup activities, and more.

CWWKB

CWWKB (angel) messages cover z/OS authorized services and the Liberty Angel process (IBM Docs).

- **CWWKB0101I**: The angel process is not available. No authorized services will be loaded. The reason code is {0}.
- **CWWKB0001I**: Stop command received for server {0}.
- **CWWKB0062E:** A second instance of an angel process attempted to start on the system where one is already active.

Coding sample to write Liberty Angel process messages to the MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="CWWKB0101I, CWWKB0001I, CWWKB0062E"/>

CWWKE

CWWKE (kernel) messages cover core kernel operations, core server startup, and runtime environment initialization (IBM Docs).

- **CWWKE0939I**: A resume request completed. (IBM Docs)
- **CWWKE1101I**: Server quiesce complete. (IBM Docs)
- **CWWKE1200W**: All threads in the Liberty default executor appear to be hung. Liberty automatically increased the number of threads from {0} to {1}. However, all threads still appear to be hung. (IBM Docs)

Coding sample to enable Liberty core kernel messages:
<zosLogging enableLogToMVS="true"
wtoMessage="CWWKE0939I, CWWKE1101I, CWWKE1200W"/>

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CWWKF

CWWKF messages cover the feature manager component for Liberty servers (IBM Docs)

- **CWWKF0011I**: The {0} server is ready to run a smarter planet.
- **CWWKF0009W**: The server has not been configured to install any features.

Coding sample to enable Liberty feature manager log messages:
<zosLogging enableLogToMVS="true"
wtoMessage="CWWKF0011I, CWWKF0009W"/>

CWWKG

CWWKG messages cover Configuration Manager component for Liberty servers (IBM Docs)

- **CWWKG0014E**: The configuration parser detected an XML syntax error while parsing the root of the configuration and the referenced configuration documents. Error: {0} File: {1} Line: {2} Column: {3}.
- **CWWKG0102I**: Multiple values are specified for the same configuration property {0}.

Coding sample to enable Liberty Configuration Manager messages:
<zosLogging enableLogToMVS="true"
wtoMessage="CWWKG0014E, CWWKG0102I"/>

CWWKO

CWWKO messages cover TCP channel and networking components for Liberty servers (IBM Docs)

- **CWWKO0219I**: TCP Channel {0} has been started and is now listening for requests on host {1} port {2}.
- **CWWKO0801E:** The SSL connection cannot be initialized from the {1} host and {2} port on the remote client to the {3} host and {4} port on the local server. Exception: {0}.
    - A new connection failed to complete a successful secure handshake. The most common reason is that the client sent an unencrypted message to a secure port. Another common reason is that an SSL certificate expired.

Coding sample to write TCP channel and networking messages to the MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="CWWKO0219I, CWWKO0801E"/>

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CWWKS

CWWKS (security) messages cover a specific range of log codes used for security, authentication, and single sign-on (SSO) components. These codes cover subsystems like OpenID Connect (OIDC), Java Web Tokens (JWT), token introspection, and user registries for Liberty servers (IBM Docs)

- **CWWKS4118E**: LTPA configuration error. A keysPassword attribute is not configured on the element, the 'ltpa_keys_password' environment variable is not set, and the 'keystore_password' environment variable is not set
- **CWWKS2907E**: SAF Service {0} did not succeed because user {4} has insufficient authority to access APPL-ID {5}. SAF return code {1}. RACF return code {2}. RACF reason code {3}.
- **CWWKS2960W**: Cannot create the default credential for SAF authorization of unauthenticated users. All authorization checks for unauthenticated users will fail. The default credential could not be created due to the following error:
- **CWWKS2907E**: SAF Service IRRSIA00_CREATE did not succeed because user WSGUEST has insufficient authority to access APPL-ID BBGZDFLT. SAF return code 0x00000008. RACF return code 0x00000008. RACF reason code 0x00000020.
- **CWWKS2930W**: A SAF authentication attempt using authorized SAF services was rejected because the server is not authorized to access the APPL-ID BBGZDFLT. Authentication will proceed using unauthorized SAF services. Since SAF authentication has failed, Liberty now attempts to use OMVS to authenticate the user's credentials, but OMVS authentication has not been configured
- **CWWKS2933E**: The username and password could not be checked because the BPX.DAEMON profile is active, and the address space is not under program control.
- **CWWKS4000E**: A configuration exception has occurred. The requested TokenService instance of type Ltpa2 could not be found.
- **CWWKS1725E:** The resource server failed to validate the access token because the validationEndpointUrl [{0}] was either not a valid URL or could not perform the validation.
- **CWWKS2908W:** SAF unauthenticated user {0} does not have the RESTRICTED attribute set.
    - The SAF user that represents the unauthenticated user does not have the RESTRICTED attribute set. The RESTRICTED attribute prevents the unauthenticated user from inheriting access to resources to which it was not explicitly granted access.

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Coding sample to write Liberty security messages to the MVS Log:
<zosLogging enableLogToMVS="true"
wtoMessage="CWWKS4118E, CWWKS2907E, CWWKS2960W, CWWKS2907E, CWWKS2930W, CWWKS2933E, CWWKS4000E, CWWKS1725E"/>

CWWKT

CWWKT messages cover HTTP transport, web container routing, and virtual host configurations for Liberty servers (IBM Docs)

- **CWWKT0016I**: Web application available ({0}): {1}. (IBM Docs)

  Coding sample to enable Liberty core kernel & runtime log messages:
  <zosLogging enableLogToMVS="true"
  wtoMessage="CWWKT0016I"/>

CWPKI

CWPKI messages in WebSphere Liberty relate to Public Key Infrastructure (PKI), SSL/TLS certificates, or keystore operations (IBM Docs)

- **CWPKI0016W**: The certificate with alias {0} from keyStore {1} will be expired in {2} days.
- **CWPKI0017E**: The certificate with alias {1} from keyStore {2} is expired.

  Coding sample to enable Liberty PKI, SSL/TLS, & Keystore operations messages:
  <zosLogging enableLogToMVS="true"
  wtoMessage="CWPKI0016W, CWPKI0017E"/>

FFDC

FFDC messages in WebSphere Liberty are automated diagnostic alerts.

- **FFDC1015I**: An FFDC Incident has been created: "{0}" at {1}. (IBM Docs)

  Coding sample to write FFDC automated alert messages to the MVS Log:
  <zosLogging enableLogToMVS="true"
  wtoMessage="FFDC1015I"/>

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